

MOSFETs: Discusión de Ejemplos 12 septiembre

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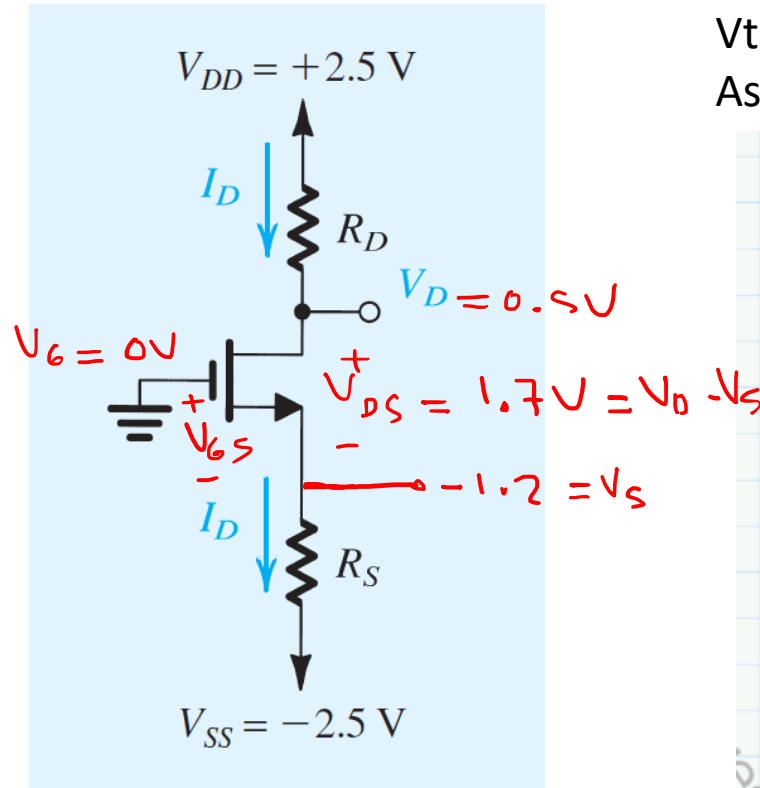
Universidad de El Salvador

Ejemplo 3

Diseñar el circuito del ejemplo3, de manera que $I_D = 0.4 \text{ mA}$,
y $V_D = +0.5 \text{ V}$.

El transistor tiene:

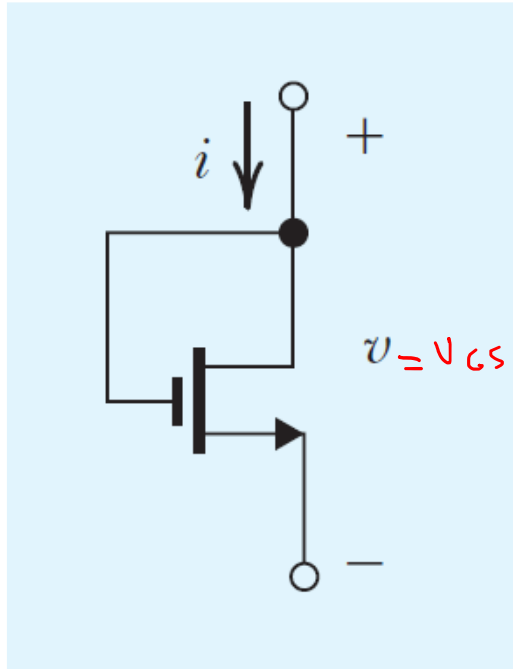
$V_t = 0.7 \text{ V}$; $\mu_n C_{ox} = 100 \text{ uA/V}^2$; $L = 1 \text{ um}$; $W = 32 \text{ um}$
Asumir $\Lambda = 0$



$$\begin{aligned}
 V_t &:= 0.7 & k'_n &:= 100 \frac{\text{uA}}{\text{V}^2} & L &:= 1 & W &:= 32 \\
 k'_n &:= \frac{k'_n}{1000} = 0.1 \\
 I_D &:= 0.4 \text{ mA} & V_D &:= 0.5 \text{ V} \\
 V_{DD} &:= 2.5 & V_{SS} &:= 2.5 \\
 R_D &:= \frac{V_{DD} - V_D}{I_D} = 5 & V_{OV} &:= \sqrt{\frac{2 \cdot I_D}{k'_n \cdot \frac{W}{L}}} = 0.5 \\
 V_{GS} &:= V_{OV} + V_t = 1.2 \\
 R_S &:= \frac{V_{SS} - V_{GS}}{I_D} = 3.25
 \end{aligned}$$

Ejemplo 4

La figura del ejemplo 4 muestra un transistor con sus terminales drenaje y compuerta conectados juntos. Encontrar la relación i-v del dispositivo resultante en términos de los parámetros del MOSFET $K_N = k'_n (W/L)$ y V_{tn} . Este dispositivo de dos terminales es conocido como transistor conectado como diodo.



$v = v_D = v_G$ garantiza saturación

$v = v_{GS} = v_{DS}$

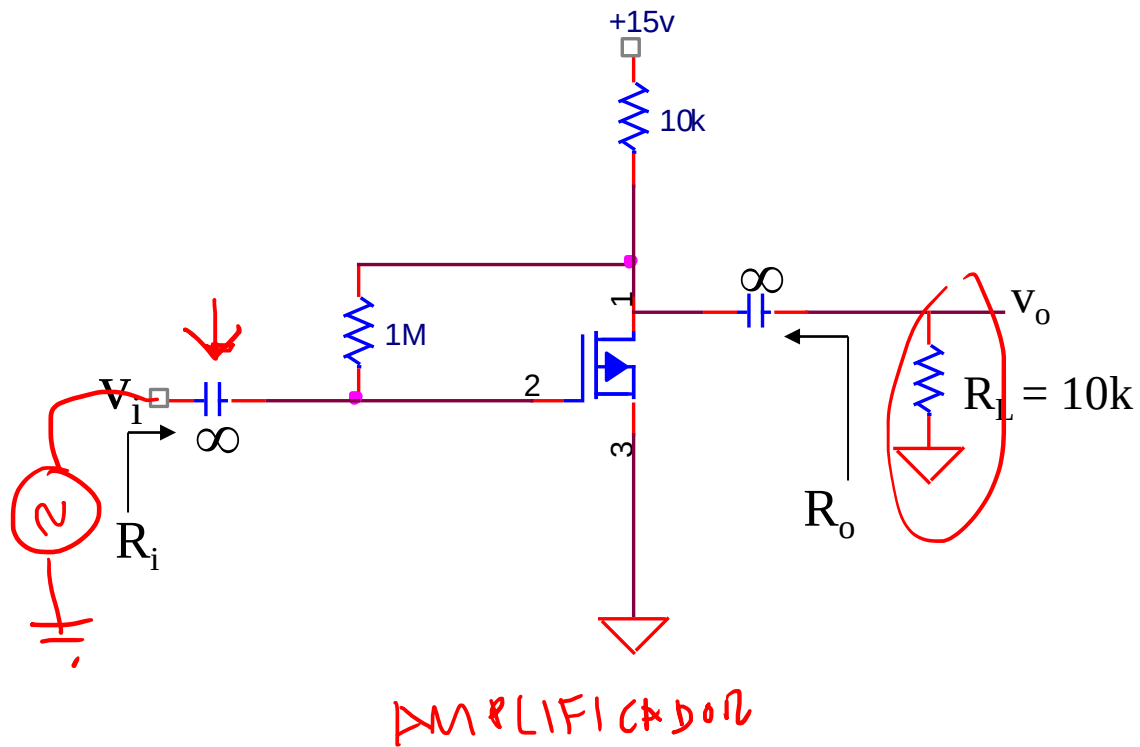
La consecuencia más importante es que el valor de resistencia viendo hacia la compuerta cambia dramáticamente

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$$

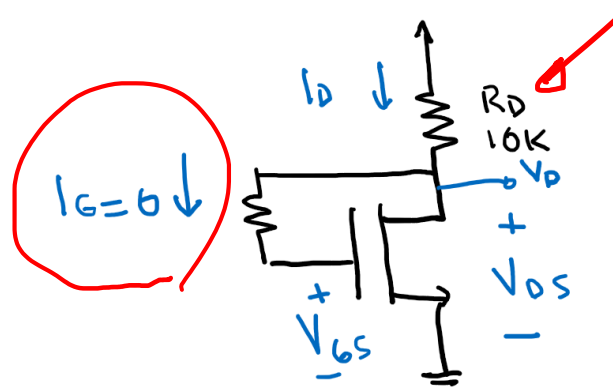
$$i = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v - V_{tn})^2$$

$$i = \frac{1}{2} k_n (v - V_{tn})^2$$

Ejemplo de aplicación



CIRCUITO EN DC
 $V_{DD} = 15V$



$V_D = V_{DS} = V_{CC}$

$$\textcircled{1} \quad I_D = \frac{1}{2} \cdot 0.25 (V_D - V_t)^2$$

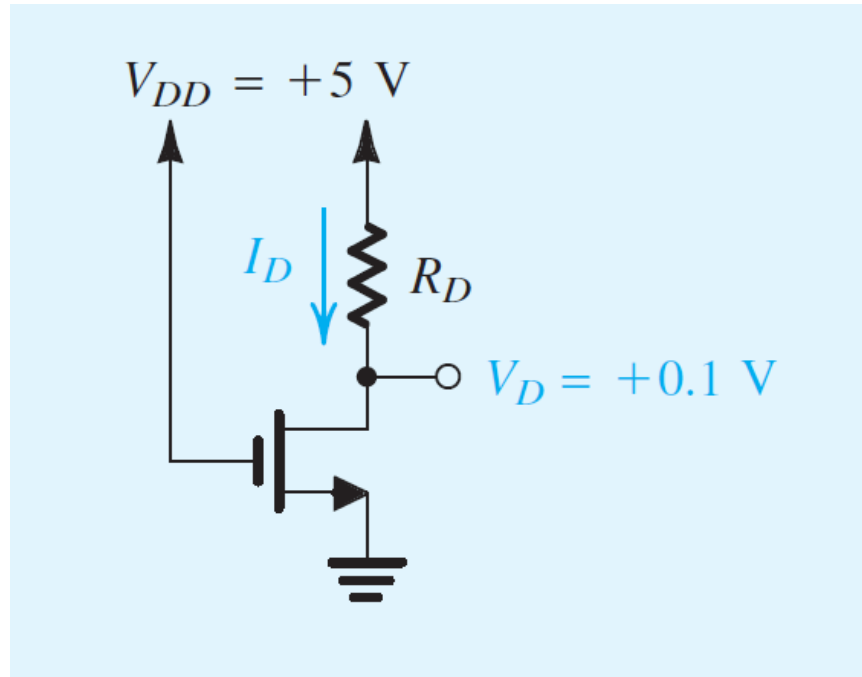
$$\textcircled{2} \quad I_D = \frac{15 - V_D}{R_D} = \frac{15 - V_D}{10}$$

$$aX^2 + bX + c = 0$$

$$R_D I_D + \sqrt{\frac{2}{k_n}} \sqrt{I_D} + (V_t - V_{DD}) = 0$$

$$a = R_D \quad b = \sqrt{\frac{2}{k_n}} \quad c = V_t - V_{DD}$$

EJEMPLO 5.



Diseñar para $V_D = 0.1\text{ V}$

$V_t = 1\text{ V}$

$k'_n / (W/L) = 1\text{ mA/V}^2$

Ejemplo_5

$V_t := 1$

$k_n := 1$

$V_{DD} := 5$

$V_{GS} := 5$

$V_D := 0.1$

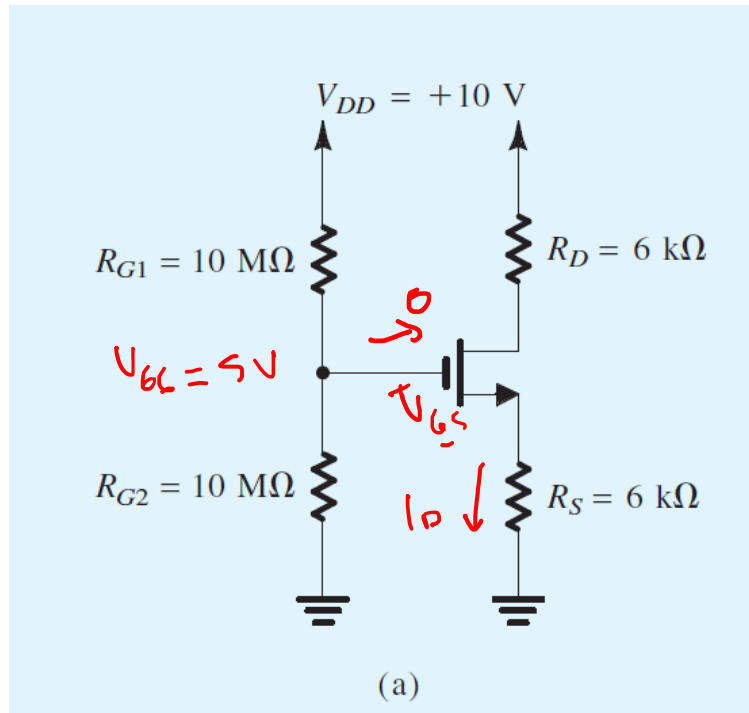
“Funcionamiento en región triodo”

$$i_D := k_n \cdot \left((V_{GS} - V_t) \cdot V_D - \frac{1}{2} \cdot V_D^2 \right) = 0.395$$

$$R_D := \frac{V_{DD} - V_D}{i_D} = 12.405 \quad \text{kohm}$$

$$r_{DS} := \frac{V_D}{i_D} = 0.253 \quad \text{kohm}$$

Ejemplo 6



Analizar el circuito del ejemplo 6, para determinar las tensiones en todos los nodos y las corrientes en las ramas.

$$V_t = 1\text{ V}; \quad \mu_n C_{ox} = k'_n = 1\text{ mA/V}^2$$

$$5\text{ V} = V_{GG} = V_{GS} + I_D R_S \quad (2)$$
$$(1) \quad I_D = \frac{1}{2} k'_n (V_{GS} - V_t)^2$$

$$RS := 6$$

$$VGG = 5$$

$$VG := 5$$

$$RS := 6$$

+

$$a := RS$$

$$b := \sqrt{\frac{2}{KN}} = 1.414$$

$$c := Vt - VGG$$

$$x1 := \frac{-b + \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a} = 0.707$$

$$ID := x1 \cdot x1 = 0.5$$

$$VOV := \sqrt{\frac{2 \cdot ID}{KN}} = 1$$

$$VGS := VOV + Vt = 2$$

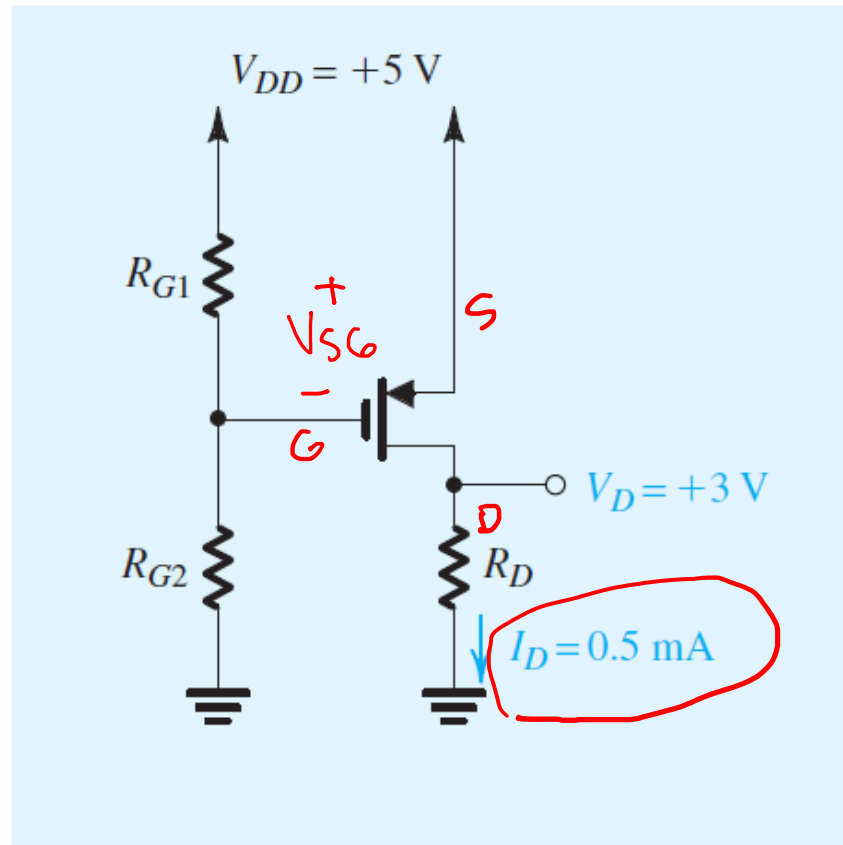
$$ID \cdot RS + VGS = 5$$

Ejemplos adicionales

Ejemplo 7

PMOS

Seleccionar R_D , R_{G1} y R_{G2}



ley
ohm

$$R_D = \frac{V_D}{I_D} = \frac{3V}{0.5mA} = 6k\Omega$$

Diseñar el circuito del ejemplo 7, de manera que el transistor funcione en saturación con $I_D = 0.5 \text{ mA}$ y $V_D = +3V$.

Hagamos que el transistor tenga:

$V_{tp} = -1V$ y $k'_p(W/L) = 1 \text{ mA/V}^2$. Asumir $\lambda = 0$.

¿Cuál es el valor mayor que puede tener R_D , al tiempo que se mantiene el funcionamiento en saturación?

$$I_D = \frac{1}{2} K_P |V_{ov}|^2 \Rightarrow |V_{ov}| = \sqrt{\frac{2I_D}{K_P}}$$

$$|V_{ov}| = V_{SG} - |V_t| \quad V_{SG} = |V_{ov}| + |V_t|$$

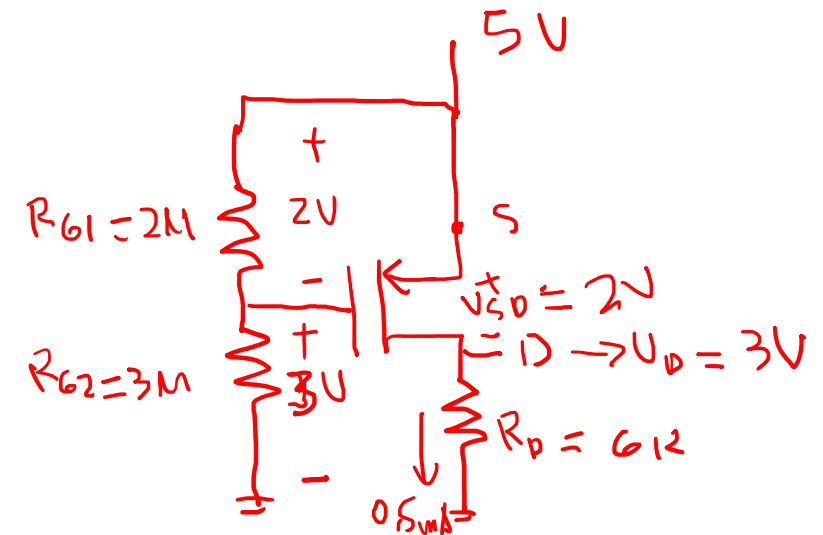
$$I_D = 0.5 \text{ mA} ; |V_t| = 1 \text{ V} ; K_P = \frac{1 \text{ mA}}{\text{V}^2}$$

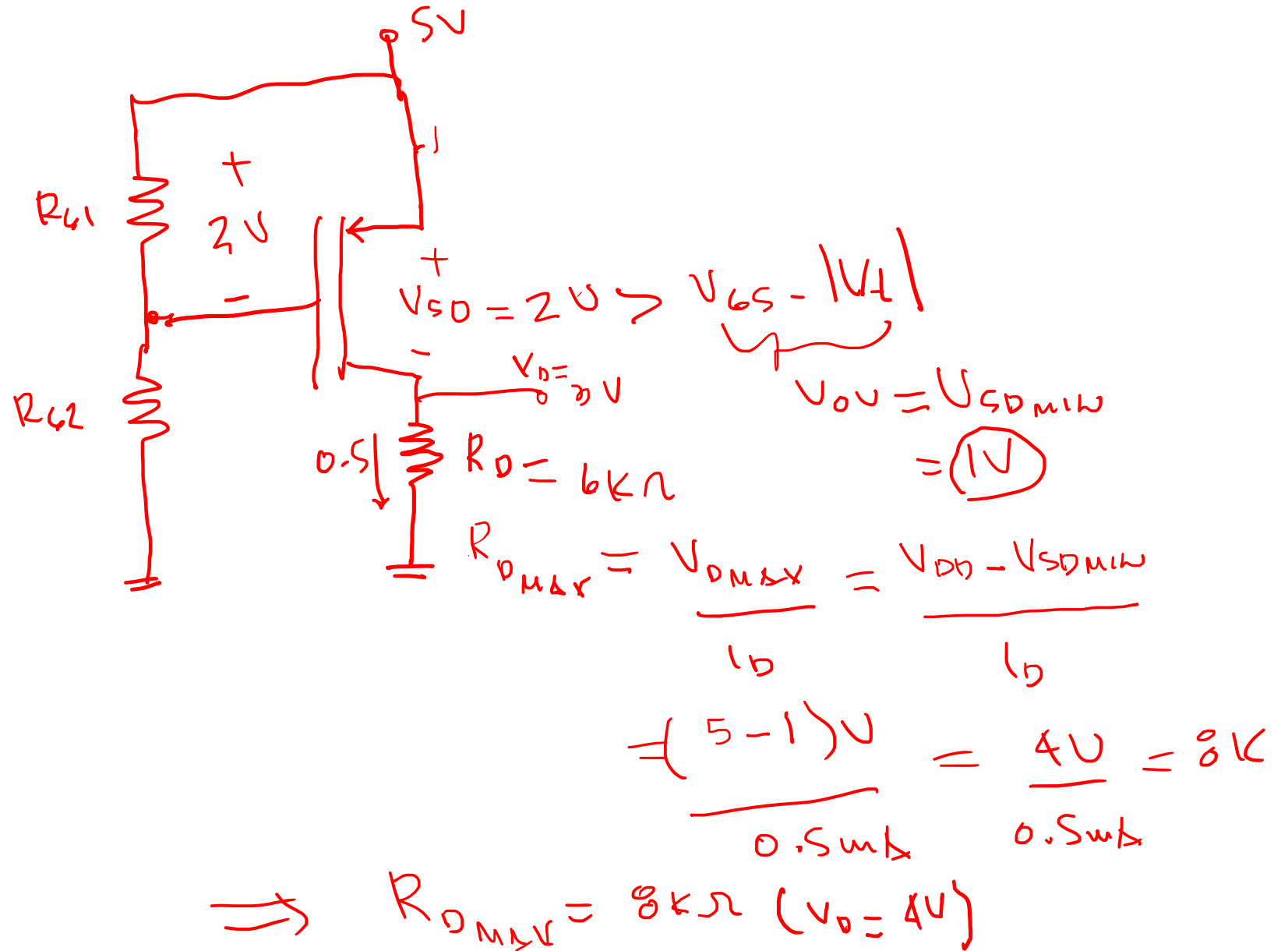
$$|V_{ov}| = \sqrt{\frac{2I_D}{K_P}} = \sqrt{\frac{2 \times 0.5 \text{ mA}}{\frac{1 \text{ mA}}{\text{V}^2}}} = 1 \text{ V}$$

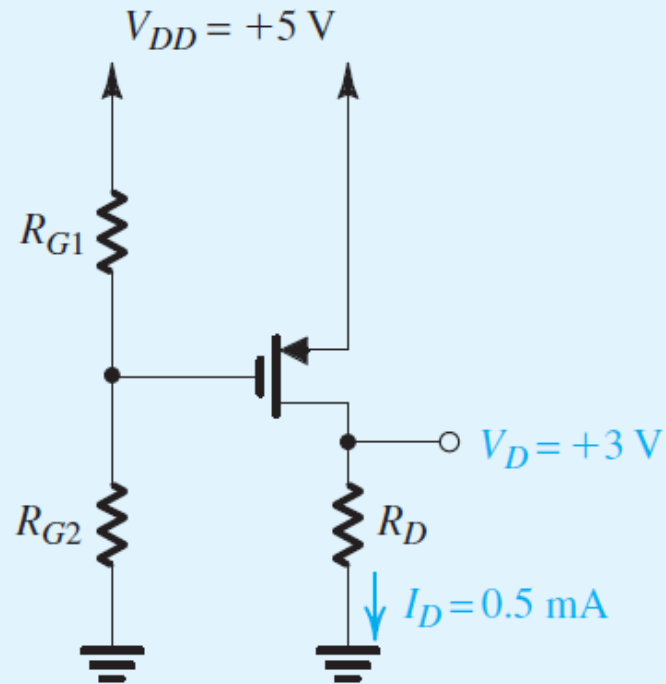
$$V_{SG} = |V_{ov}| + |V_t| = 2 \text{ V}$$

$$I_D = \frac{1}{2} K_P [V_{GS} - V_t]^2$$

$$= \frac{1}{2} K_P [V_{SG} - |V_t|]^2 = \frac{1}{2} K_P |V_{ov}|^2$$







$$V_t := 1 \quad K_p := 1 \quad I_D := 0.5 \quad V_D := 3$$

$$R_D := \frac{V_D}{I_D} = 6 \quad V_{OV} := \sqrt{\frac{2 \cdot I_D}{K_p}} = 1$$

$$V_{SG} := V_{OV} + V_t = 2$$

$$V_{DD} := 5$$

$$V_G := V_{DD} - V_{SG} = 3$$

“La solución para R_{G2} , R_{G2} no es única”

$$R_{G2} := 300 \quad k\Omega$$

$$R_{G1} := 200 \quad k\Omega$$

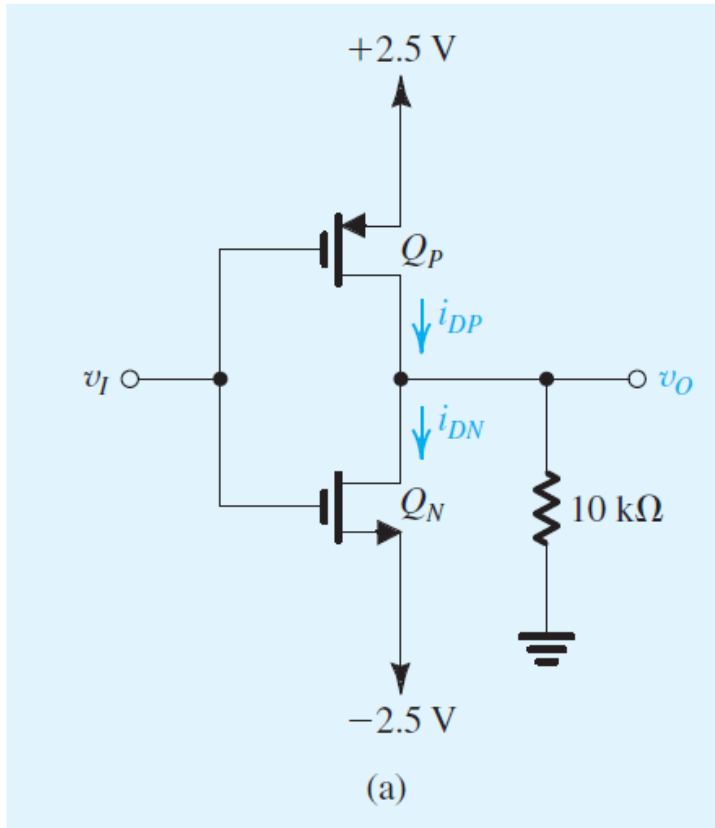
$$V_G := V_{DD} \cdot \frac{R_{G2}}{R_{G2} + R_{G1}} = 3$$

“Valor máximo de R_D ”

$$V_{DSmin} := V_{OV} = 1 \quad V_{Dmax} := V_{DD} - V_{DSmin} = 4$$

$$R_{Dmax} := \frac{V_{Dmax}}{I_D} = 8$$

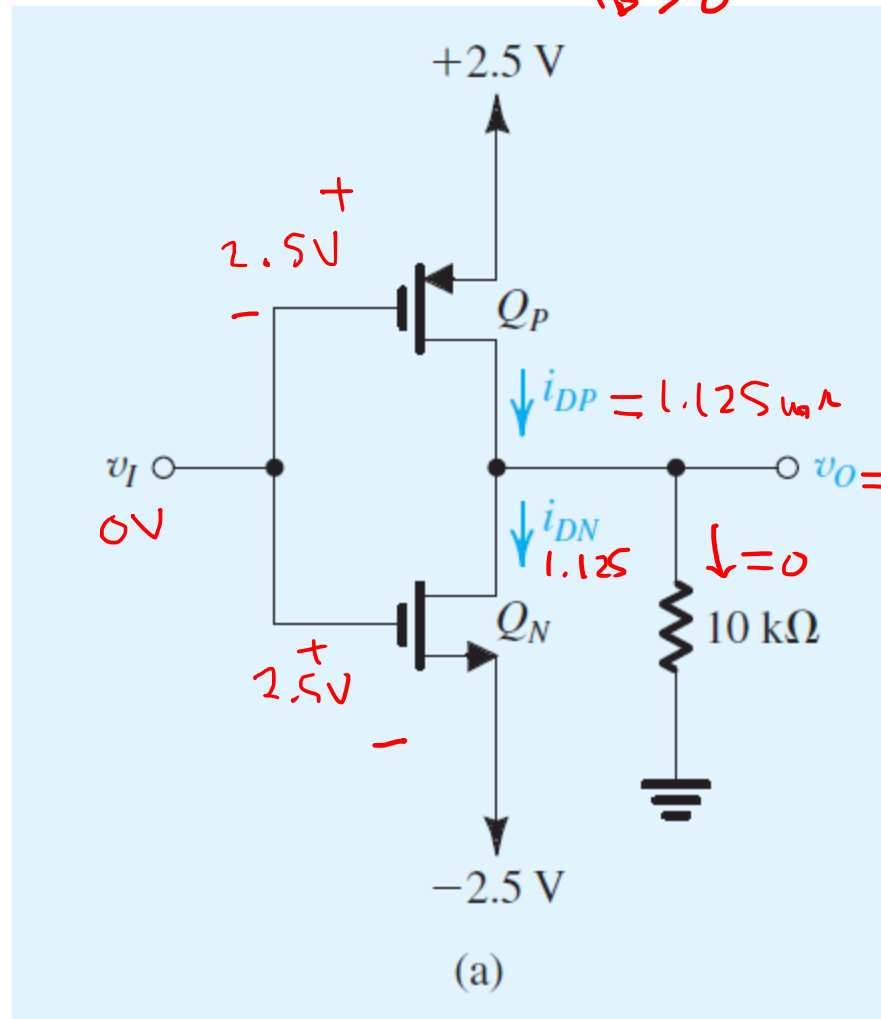
Ejemplo 8



Circuito para el ejemplo 8

Los transistores NMOS y PMOS del circuito, son coincidentes, con $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1 \text{ mA/V}^2$ y $V_{tn} = -V_{tp} = 1 \text{ V}$. Asumir $\Lambda = 0$ para ambos dispositivos. Encontrar las corrientes de drenaje i_{DN} e i_{DP} , también como la tensión v_O , para $v_i = 0, 2.5 \text{ V}$ y -2.5 V .

$$|V_{GS}| > |V_t| \Rightarrow \begin{matrix} \text{EXISTE} \\ \text{CANAL} \\ I_D > 0 \end{matrix}$$



$$|V_t| = 1V \quad |K| = 1 \frac{\text{mA}}{\text{V}^2}$$

$$\text{con } V_i = 0$$

$$|V_{GS}| = 2.5V \text{ para ambos transistores}$$

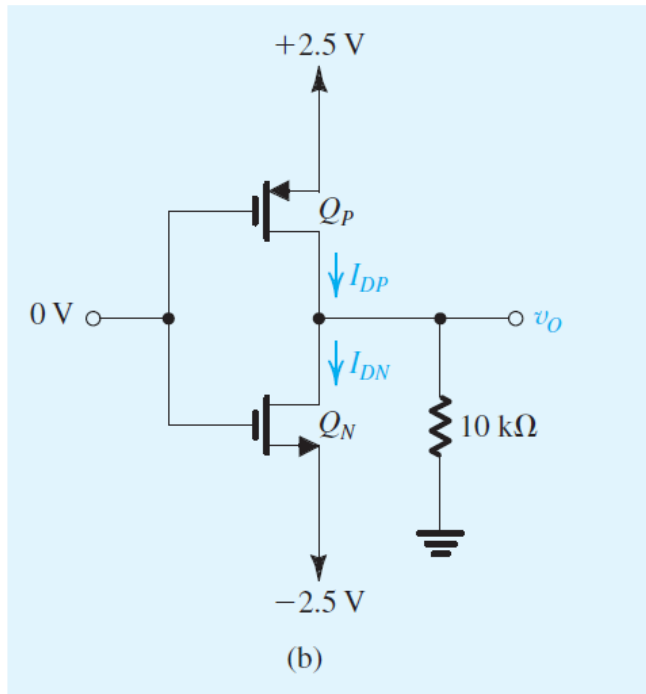
$$I_{DP} = I_{NP} = \frac{1}{2} K (V_{GS} - V_t)^2$$

$$= \frac{1}{2} \times 1 \frac{\text{mA}}{\text{V}^2} (1.5V)^2$$

$$= 0.5 \frac{\text{mA}}{\text{V}^2} \times 2.25V^2$$

$$= 1.125 \text{ mA}$$

$$I_{DN} = I_{DP} = 1.125 \text{ mA}$$



Circuito con $v_i = 0V$

Ya que los transistores son coincidentes, $v_O = 0$
 y $v_{DSp} = v_{SDp} = 2.5 V$, lo que garantiza
 saturación, ya que $v_{GSn} = v_{SGp} = 2.5V$

Ejemplo_8

$K := 1$

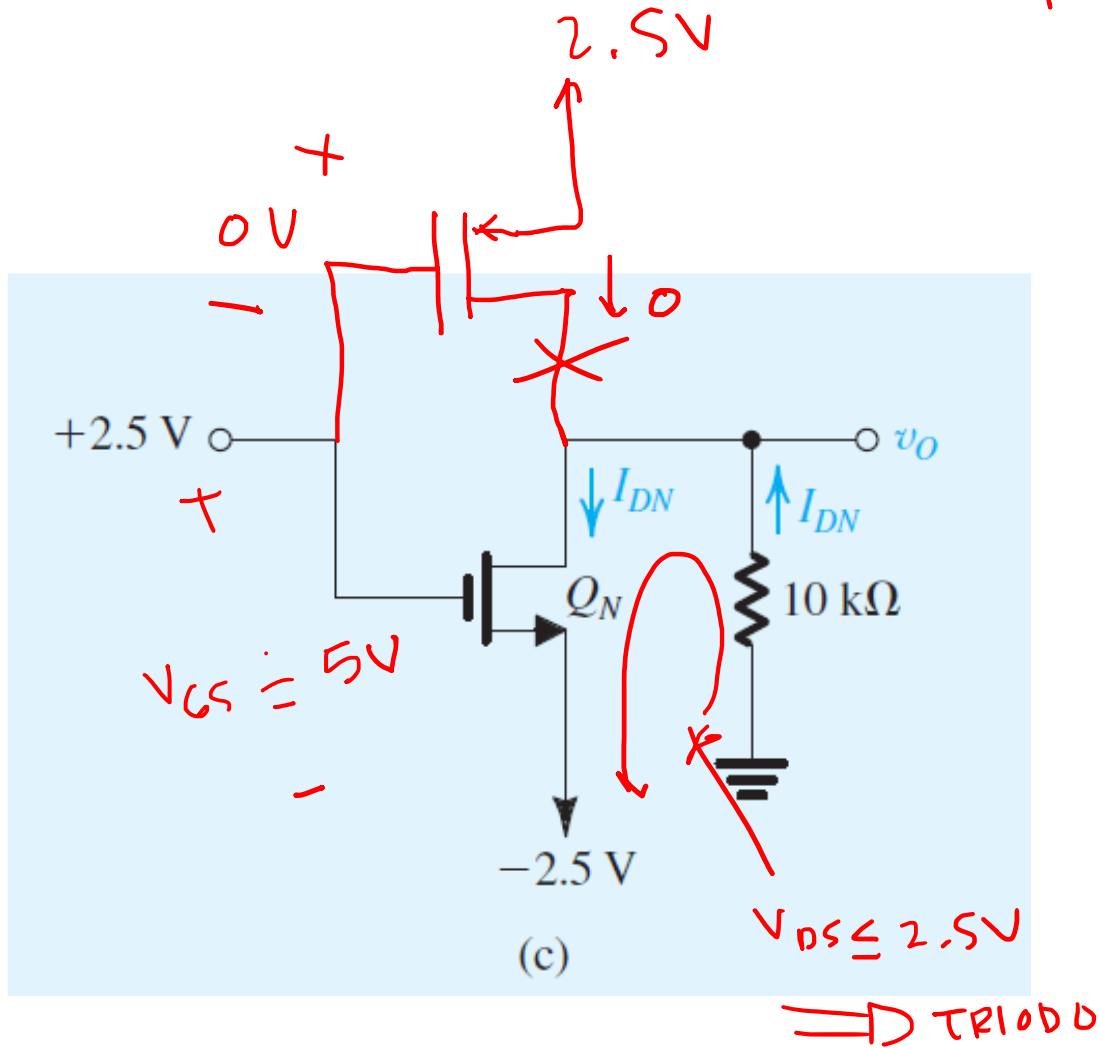
$Vt := 1$

Literal_a

+

$VGS := 2.5$

$$iD := \frac{1}{2} \cdot K \cdot (VGS - Vt)^2 = 1.125$$



para saturación
 $v_{DS} \geq v_{GS} - V_t$
 $v_{DS} \geq 4V$

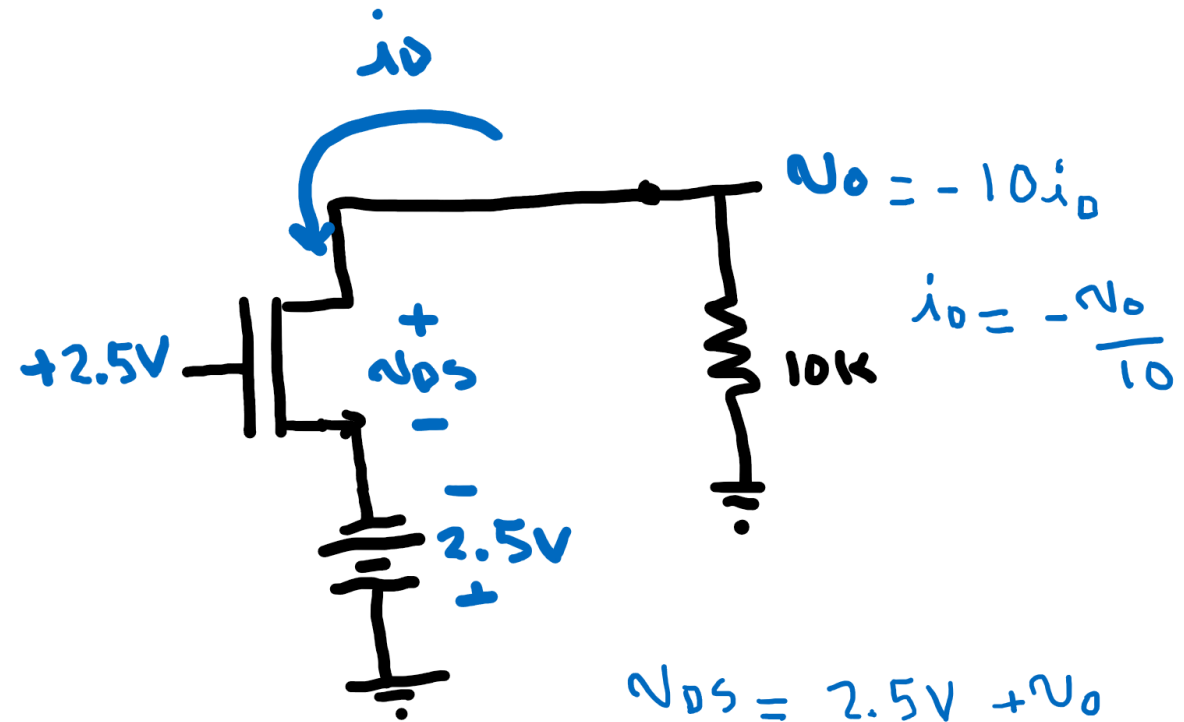
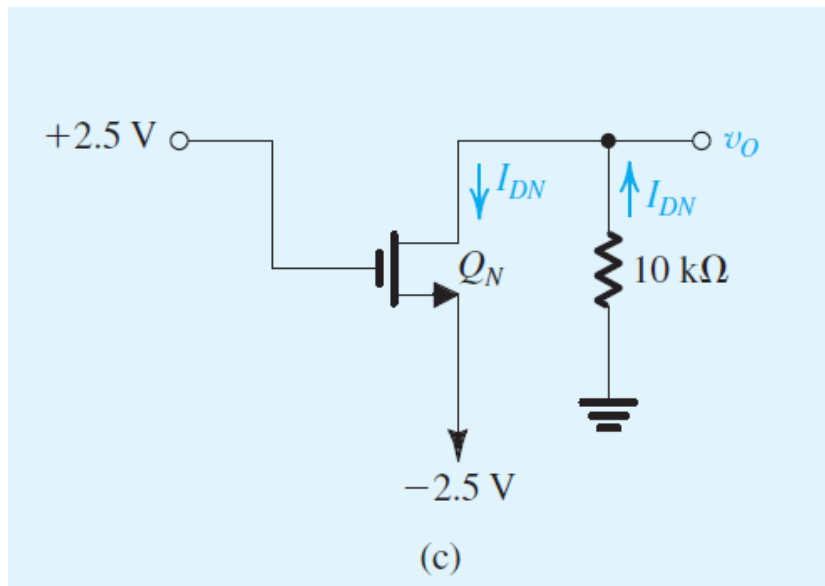
Con $v_I = 2.5$, $v_{SG} = 0$ para el PMOS y se apaga (cut off), de manera que solo el NMOS interviene en el circuito.

$v_{GS} = 5V$ para el NMOS

Y $v_O = v_D$ es negativo

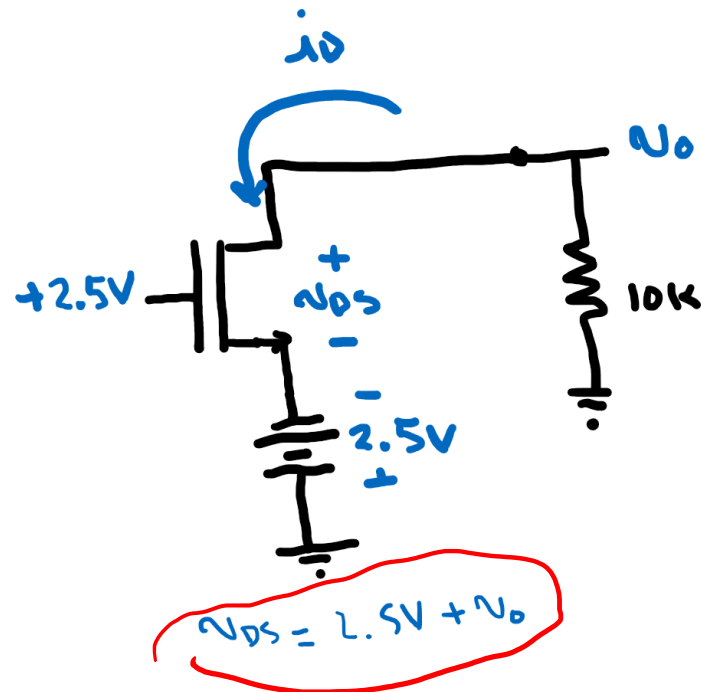
Con lo que $v_{DS} < 2.5$, es decir $v_{DS} < V_{OV}$
 Por lo tanto el NMOS $\rightarrow$ triodo

Además asumimos v_{DS} muy pequeño, con lo que se puede simplificar la ecuación de i_D



$$i_D = K_n \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

si v_{DS} pequeño $\Rightarrow$ despreciamos v_{DS}^2



$$i_D = -\frac{v_O}{10}$$

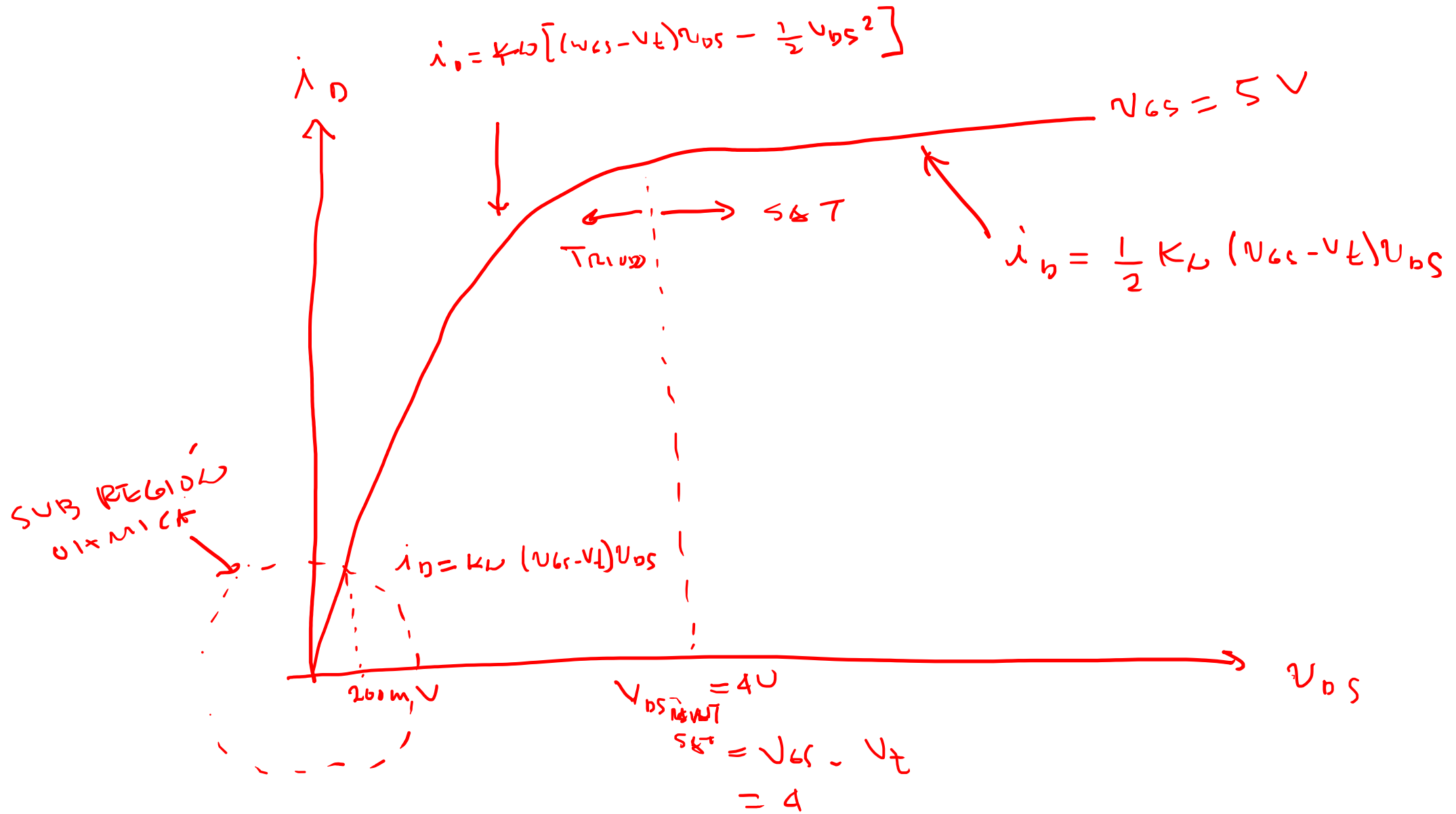
$$i_D = K_n (v_{GS} - V_t) v_{DS} = -v_O / 10$$

$$= K_n (5 - 1) (2.5 + v_O)$$

$$\frac{v_O}{10} = 4 (2.5 + v_O)$$

$$v_O = -2.44V \quad i_D = 0.244mA$$

Con $v_O = -2.44V$, se tiene $v_{DS} = 0.06V$ (muy pequeño), lo cual confirma que el uso de la expresión simplificada para i_D es adecuado.



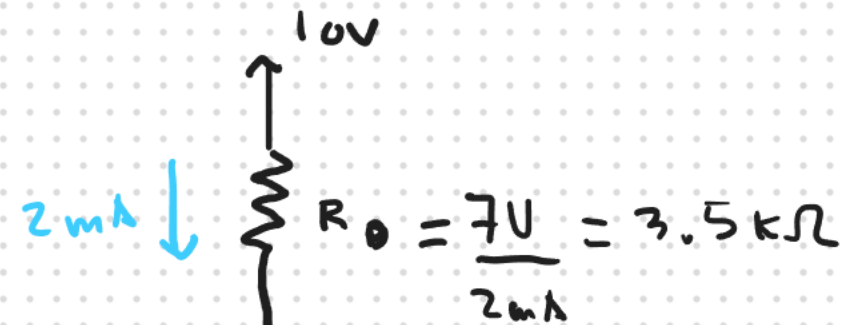
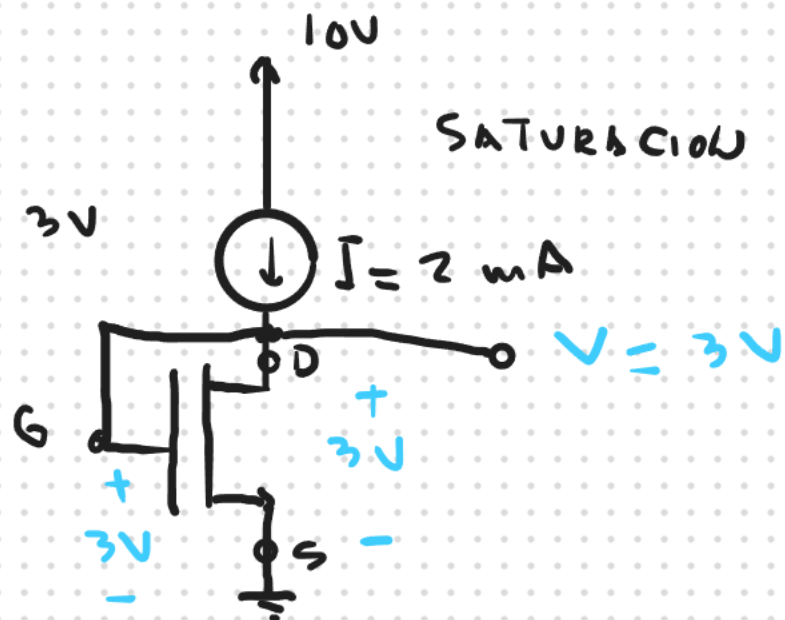
P 33

$$V_{GS} = V_{DS} = 3V$$

$$V_t = 1V$$

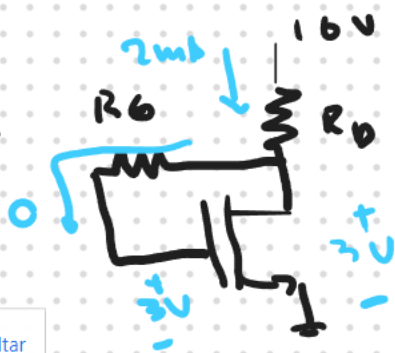
$$I_D = \frac{1}{2} k_N (V_{GS} - V_t)^2$$

$$k_N = \frac{2I_D}{V_{GS}^2} = \frac{2 \times 2mA}{(2V)^2} = \frac{4mA}{4V^2} = 1 \frac{mA}{V^2}$$



$$R_D = \frac{V_{DD} - V_{DS}}{I_D} = \frac{7V}{2mA} = 3.5k\Omega$$

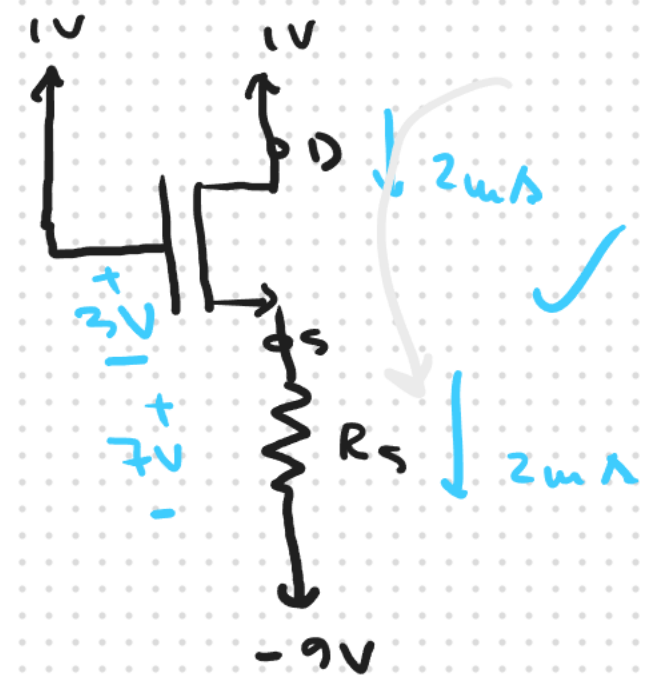
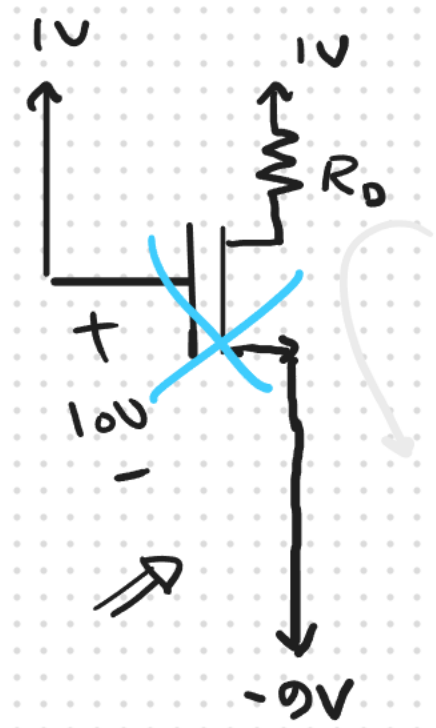
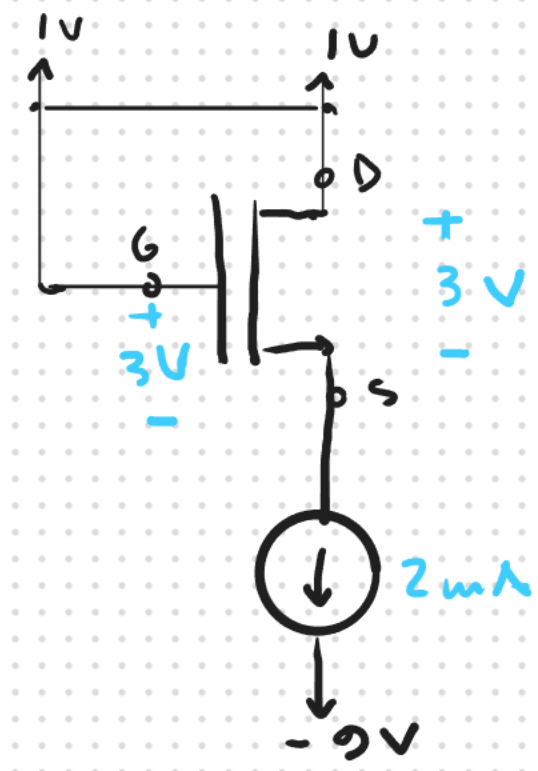
$\{ R_{Gmax} ? \}$



P 33 (b)

$$V_t = 1V$$

$$K_n = 1 \text{ mA/V}^2$$



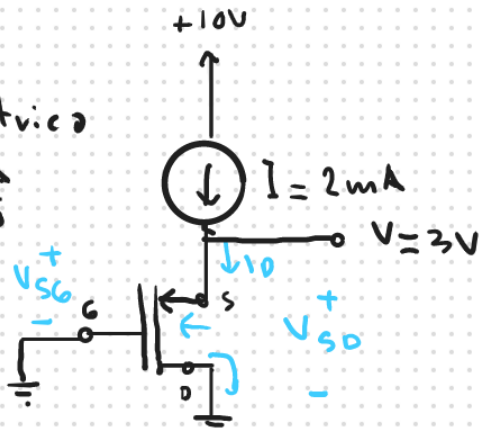
$$R_S = \frac{7V}{2mA} = 3.5 \text{ k}\Omega$$

P33(c)

Transistor simétrico

$$K_p = K_n = 1 \frac{\text{mA}}{\text{V}^2}$$

$$|V_t| = 1\text{V}$$

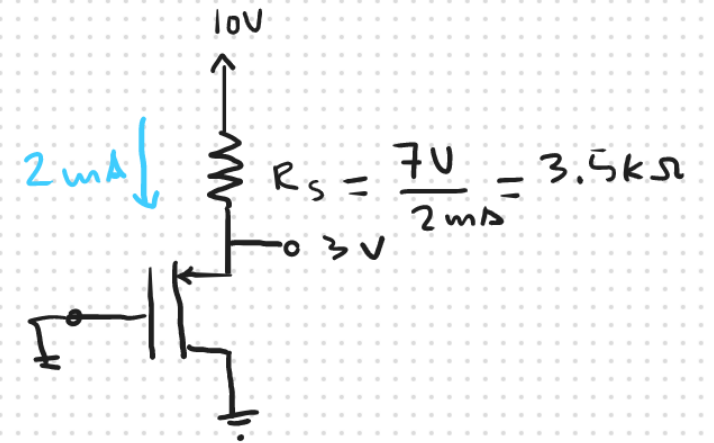


$$V_{SG} = V_{SD} = 3\text{V}$$

$$I_{DP} = \frac{1}{2} K_p (V_{SG} - |V_t|)^2$$

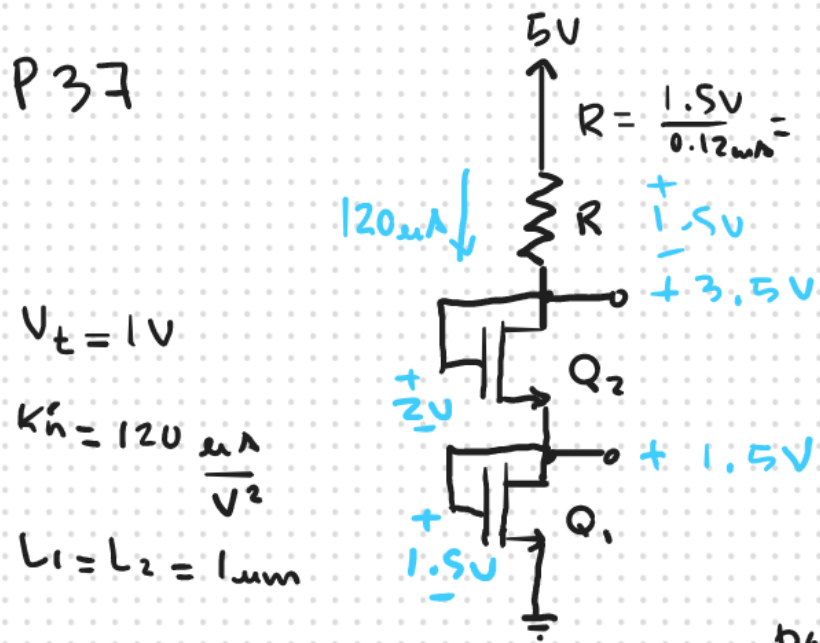
$$|V_{ov}| = \sqrt{\frac{2I_{DP}}{K_p}} = \sqrt{\frac{2 \times 2\text{mA}}{1 \frac{\text{mA}}{\text{V}^2}}} = \sqrt{4\text{V}^2} = 2\text{V}$$

$$V_{SG} = |V_{ov}| + |V_t| = 2\text{V} + 1\text{V} = 3\text{V}$$



$$R_S = \frac{7\text{V}}{2\text{mA}} = 3.5\text{k}\Omega$$

P37



$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$\left\{ \begin{array}{l} W_1? \\ W_2? \end{array} \right.$$

$$I_{D1} = 120 \mu A \quad V_{GS1} = 1.5V$$

$$I_{D2} = 120 \mu A \quad V_{GS2} = 2V$$

para Q1

$$I_{D1} = \frac{1}{2} k_n' \left(\frac{W_1}{L_1} \right) V_{OV1}^2$$

para Q2

$$I_{D2} = \frac{1}{2} k_n' \left(\frac{W_2}{L_2} \right) V_{OV2}^2$$

$$\frac{W_2}{L_1} = \frac{2I_{D2}}{k_n' V_{OV2}^2} = \frac{2 \times 120 \mu A}{120 \mu A/V^2 (1V)^2}$$

$$\frac{W_2}{L_2} = 2 \Rightarrow W_2 = 2 \mu m$$

$$V_{OV1} = V_{GS1} - V_t = 1.5V - 1V = 0.5V$$

$$V_{OV2} = V_{GS2} - V_t = 2V - 1V = 1V$$

$$\text{si } L_1 = 1 \mu m \Rightarrow \boxed{W_1 = 8 \mu m}$$

$$\frac{W_1}{L_1} = \frac{2I_{D1}}{k_n' V_{OV1}^2} = \frac{2 \times 120 \mu A}{120 \mu A/V^2 \times (0.5V)^2} = \frac{2}{0.25} = 8$$

P4A_u

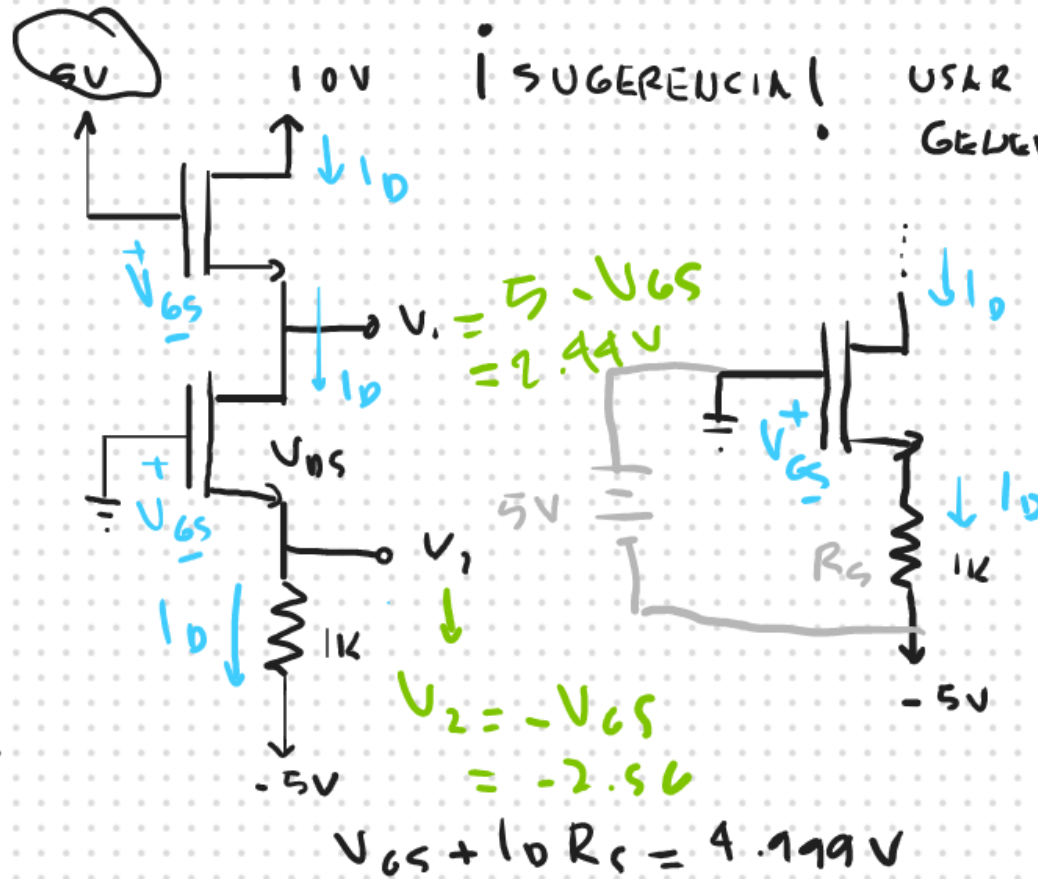
$$V_t = 1V$$

$$K_N = 2 \frac{\text{mA}}{\text{V}^2}$$

$$I_D = 2.430 \text{ mA}$$

$$V_{ov} = 1.56$$

$$V_{GS} = V_{ov} + V_t = 2.56$$



$$① I_D = \frac{1}{2} K_N (V_{GS} - V_t)^2$$

$$② V_{GG} = 5V = V_{GS} + I_D R_S$$

$$0 = R_S I_D + \sqrt{\frac{2}{K_N}} \sqrt{I_D} + (V_t - V_{GG})$$

$$a = R_S$$

$$b = \sqrt{\frac{2}{K_N}}$$

$$c = V_t - V_{GG}$$

Referencia

- Sedra. Microelectronic Circuits, 7th Edition. Oxford University Press.